

INDIRAKUMAR RAJENDIRAN

Senior Software Engineer / Technical Lead

Chicago, IL | +1 224-5844-749 | indirakumarus710@gmail.com

PROFESSIONAL SUMMARY

Senior Software Engineer and Technical Lead with 12+ years of experience designing, building, modernizing, and supporting enterprise applications across healthcare, banking, logistics, and telecom domains. Strong backend engineering expertise in C#, .NET Core, ASP.NET Web API, SQL Server, MongoDB, messaging systems, CI/CD, secure API design, and production support. Experienced in leading engineering teams, driving architecture discussions, performing code reviews, validating non-functional requirements, modernizing legacy platforms, and delivering scalable, maintainable, and compliant software solutions.

TECHNICAL SKILLS

Languages: C#, SQL, JavaScript, HTML, CSS; exposure to Spring Boot

Backend Engineering: .NET Core, ASP.NET Core Web API, ASP.NET MVC, WCF, REST APIs, Web Services, Entity Framework, LINQ

Frontend: React, jQuery, HTML/CSS

Databases and Caching: Microsoft SQL Server, MongoDB, Redis

Cloud and DevOps: Azure, AWS EC2, Jenkins, UCD, Git, GitHub, TFS, SonarQube, Swagger

Testing and Quality: MSTest, NUnit, unit testing, integration testing, static code analysis, code reviews

Architecture and Platform: API design, distributed systems, microservices, design patterns, OAuth2, JWT, observability, CI/CD, performance, scalability, reliability

Messaging and Integration: IBM MQ, MSMQ, SendGrid, iText, XML, JSON

PROFESSIONAL EXPERIENCE

Technical Lead / Senior Software Engineer | Cognizant Technology Solutions | Dec 2018 - Oct 2024

Role progression: Associate (Dec 2018 - Jul 2022), Software Engineer (Jul 2022 - Nov 2022), Sr. Associate - Projects (Dec 2022 - Jun 2023), Technical Lead (Jun 2023 - Oct 2024)

- Led a compliance engineering team of 15 Engineers delivering medical benefits management workflow services for health insurance; owned requirements analysis, technical design, release delivery, stakeholder communication, and production support.
- Designed, built, tested, and shipped enterprise .NET services end-to-end using C#, .NET Core, ASP.NET Web API, SQL Server, MongoDB, MSTest/NUnit, Jenkins, UCD, Azure, and AWS deployment workflows.
- Designed a modernization framework to migrate an AS400 logistics rate management system to .NET Core Web API, React UI, and Microsoft SQL Server, enabling API-first quote generation and reducing dependency on legacy workflows.
- Contributed to architecture roadmaps across business, application, data, and technology domains; produced high-level design proposals, proof of concept prototypes, reusable code standards, and implementation documentation.
- Performed design reviews and code reviews to validate non-functional requirements, including security, performance, maintainability, scalability, usability, and reliability.
- Implemented OAuth2/JWT authentication, structured logging, telemetry, and standardized API error handling to improve traceability, production debugging, and audit readiness.
- Improved CI/CD and engineering quality practices using Jenkins, Git, SonarQube, MSTest/NUnit, and automated deployment workflows for daily lower environment deployments and monthly Production environment deployments; supported continuous improvement by reducing technical debt and improving release quality by improving SonarQube code quality coverage to more than 90%.
- Built 15 backend APIs and database integrations for student loan workflows, including loan-detail visibility, due-balance tracking, due-date reminders, and new-loan application features.
- Designed healthcare AI proof-of-concept capabilities for claims fraud detection and explainability using C#/.NET integration patterns, SHAP, and Fairlearn, with a focus on auditability, responsible AI, HIPAA/CMS-aligned compliance considerations, and secure handling of sensitive workflows.

Analyst Programmer / Software Engineer | Ramco Systems | Mar 2017 - Dec 2018

- Developed and enhanced ERP product modules for a banking client, including journal creation, voucher processing, depreciation workflows, GST taxation, and reporting capabilities.
- Built a bulk transaction processing application using IBM MQ to support credit/debit nullification and batch NEFT transaction processing.
- Developed SQL Server backend logic, stored procedures, and Crystal Reports for monthly banking transaction reporting and operational review.
- Integrated SendGrid for marketing email workflows and iText for agreement-generation automation across multiple agreement formats.

Junior Executive / Backend Developer | Tata Communications Transformation Services | Feb 2014 - Mar 2017

- Developed backend components for a telecom order management system supporting account management, circuit management, service design, and pricing workflows.
- Built and maintained order-processing modules, validation logic, backend services, and operational support features in collaboration with business and delivery teams.

EDUCATION

Bachelor of Engineering in Electronics and Communication Engineering | Maamallan Institute of Technology | Sep 2009 - May 2013